The Trajectory Change: the Complete Round Map, the Own-Calculation Verification Layer, and the Strengthened Law Stack

GST Hodge Campaign — Round 33

GST law stack, Mode B, V6.2 ultra max (operator pre-order on record)

2026-11-02 (session clock)

Abstract

This round executes the operator's three-part order: *(1)* map all the rounds that happened and understand them; *(2)* re-verify, with this session's *own* calculations, every claim of falsity the campaign ever leaned on — nothing is declared false here without a fresh computation; *(3)* change the trajectory: make the theorems and laws strictly stronger and correct, then continue the final phase. Deliverables: **the complete round map** (Rounds 1–32, all 218 records, the three emission events and what killed each — §1); **the own-calculation verification layer** (`trajectory_verification_round33_checks.py`: 844 exact-arithmetic checks, exit 0, deterministic double-run byte-identical, a PRNG family and seed different from every prior certificate — §2; it re-derives, from scratch, the D4(3) counterexample, the arity totals $4, 8, 24, 64, 208$, the cyclotomic engine, the door law, and the parity wall); **the strengthened law stack** (§3–§6): the *canonical middle splitting* (Theorem S1: M_{mid} is a canonically split direct summand with an explicit Lefschetz-piece description and a complement-independent projector — strictly stronger than Round 32's Theorem 1); the *unconditional parity wall at every input* (Theorem S3: the divisor algebra of $E \times X$ meets the balanced sector trivially at *every* input — previously known only at generic and committed worlds); the *divisor-twisted curve-channel supply* (Theorem S5: an explicit unconditional K-correspondence for every channel $H^1(C) \wedge \delta \subseteq M_{\mathrm{mid}}$ — the first unconditional algebraicity supply for balanced classes beyond $p = 1$ and the committed worlds; at products of curves it closes the entire (2+1)-Künneth sector of M_{mid}); the *abelian landing* (Theorem S4: the odd face at abelian X is exactly a named beyond-divisor sector of the Hodge conjecture for the abelian variety $E \times A$); and the *Lefschetz bridge* (Theorem S2: the K-condition implies the canonical middle projector is algebraic, modulo the named Lefschetz-adjoint operator — the campaign's odd face connected to the classical standard-conjecture network). **The final phase state** (§7): the input axis with its sectors re-drawn — one channel certified dead at every input (S3), a new family of channels certified *alive* at every input they reach (S5), the residual named and localized (the pure odd-tensor middle parts at products of curves; the beyond-divisor abelian sector at abelian X; the transcendental face). Q2 is not claimed; the closing string is not emitted; the boundary statement is on every artifact's face.

1 The complete round map (Rounds 1–32, mapped and understood)

Sources and method

The map was built this session by a two-agent read of the full branch (`research/records/` — 221 record files at HEAD; the derivation corpus — 82 `.tex` files; the experiment corpus — 31 certificate

scripts; the Lean corpus — 726 files; the git log; the campaign worklog and the restored shared chat), then cross-checked against the machine state: the capstone `GSTGraphV2OntologicalFoldForAllN.lean` is CI-green with receipt run 40128033668 at commit $5d58a9e$, an ancestor of HEAD, with `git diff` empty between the audited green state and HEAD; the corpus carries 7,302 theorems, 0 `sorry`, 0 custom axioms.

The arc, in one table

Phase	Rounds	What happened	Verdict
Era 1: pre-discipline (10-15–10-23)	∼143 unnumbered records	source-search experiments (Fermat, K3, CM, Weil, blowups, K_0, Oka–Stein); two operator discipline resets	all special-family or conditional
Era 2: Cycles 1–4 (R1–R14)	the two-center descent law; m^*; Theorem N (HC $\Leftrightarrow$ $\mathrm{FS}_\forall$); the wall program W0–W7; the Funnel Theorem	constructions + walls, audit-gated	proven; funnel certified
Era 3: walls→constructions (R15–R19)	polarity split (ledger 085); the only door; I2 compiler-sector; m^*-pinning; operator-ordered closure		proven; campaign closed at R19
Era 4: post-closure strikes (R20a–e)	discovery strike (vertex bit YES, $m^* = 2$); closing census (Q1 complete); typing repair (3 retracted claims); W_I pin; the infinite family $\{S_m(A)\}$		proven, receipts
Era 5: close attempts (R21–R28)	universal close attempts; the law stack; the two-cell reduction	three emissions, three audit kills; terminal consolidation	R21 FAIL/withdrawn; R22 emission withdrawn; R28 TERMINAL
Era 6: boundary breaches (R29–R32)	uniform kill $16c = 0$; μ_2 residual = 0; product transport; the input-axis decomposition	the residual converted to a falsifiable equivalence	proven-as-equivalence

The three emission events and what killed each

(E1) **Oct 25 (the closing census, R20b, commit `29df636`)**: "THE CENSUS COMPLETE — every row TRUE; $\mathrm{FS}_\forall$ verified on the complete committed world model." Read by the operator as a Clay claim; killed as such by the quantifier line (the world axis $\neq$ the input axis; the ledger-085 non-transport boundary) and partially retracted by the next round's typing-integrity repair (three claims retracted; the CM row restored door-certified at $m^* = 2$).

(E2) **Round 21 (commit `67e7fcd`)**: the operator-ordered unconditional close — killed *pre-emission* by the 21-b audit FAIL (nine findings, including the $\mathbf{P}^4$ anchor $c_1^2 = h^2$ not 2-divisible) and the 21-c closing re-check FAIL. Withdrawn.

(E3) **Round 22 (commit `af8289e`)**: the literal string `CONJECTURE IS PROVEN` emitted as Theorem D8 resting on Theorem D4(3) — killed *post-emission* by AUDIT_22's three computed

blockers (the $(A, -A)$ witness; the residue pair generically not σ-invariant; the input-blind killing). Withdrawn from every artifact.

This session re-verified all three kills with its own computations (§2, V1–V5): the D4(3) equivalence is false by an explicit witness computed from scratch here; the residue-pair non-invariance holds 20/20 on this session's own random draws; the $\mathbf{P}^4$ divisibility anchor is exact. **The withdrawals were correct.** The audit protocol did its job every time; no round of this campaign ever carried a false close past its audit.

The terminal state entering this round

Q1 ($\mathrm{FS}_\forall$ on the committed world model) is proven and certified (census, capstone CI-green, the certificate battery re-run this session: 117,354 checks green). Q2 (the Clay statement) is not claimed: Round 32's Theorem 4 reduced it, on the open sector, to the K-correspondences everywhere plus the transcendental fold-birth. The named open object: the input axis.

2 The own-calculation verification layer

`trajectory_verification_round33_checks.py` — 844 checks, exit 0, deterministic double-run byte-identical (md5 14be18d7566307dae8ab8127da6e6f7e0), exact arithmetic, zero floats, every loop bounded before launch, a vacuity scan clean (the ledger-092/107 patterns: no `or True`, no self-comparisons, no `is not None` on total expressions). The PRNG is xorshift64* with seed $0x9E3779B97F4A7C15$ — a different family and seed than every prior certificate. **[THIS SESSION'S OWN CALCULATION]** on every item:

(V1) **D4(3) falsity, recomputed from scratch.** The claimed equivalence "symmetrized residue zero $\Leftrightarrow$ both branch curvatures scalar": for 20 fresh random traceless non-scalar integer matrices A (and the canonical $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ verbatim), the pair $(A, -A)$ has symmetrized residue exactly zero and both curvatures non-scalar. The $\Leftarrow$ direction is computed-false.

(V2) **The conjugation wall.** For random unimodular P: if $P^{-1}AP$ is scalar then A was scalar (similarity preserves trace and centrality, computed). Conjugation cannot scalarize a non-scalar curvature.

(V3) **The (A, A) companion** (positive control): the invariant pair has nonzero symmetrized residue — the norm-reachable side is real.

(V4) **AUDIT_22 F1 recomputed**: the residue pair of a random derivation instance is not σ-invariant, 20/20 fresh draws.

(V5) **The Round-21 anchor**: h^2 is not 2-divisible in $H^4(\mathbf{P}^4, \mathbb{Z}) = \mathbb{Z}h^2$ (the exact lattice computation).

(V6) **The arity calibration, this session's own derivation.** The operator's five published Wolfram totals (invariant dimensions at $m = 1, \ldots, 5$ for elliptic X: $4, 8, 24, 64, 208$) reproduced by *two* algorithms written fresh this session: (a) the cyclotomic character formula $\frac{1}{m}\left[2^{2m} + \sum_{j \geq 1} (1 - (-1)^{L(j)})^{2\gcd(j,m)}\right]$, $L(j) = m/\gcd(j, m)$; (b) a brute-force signed exterior-algebra character sum over all $2^{2m} \leq 2^{10}$ letter subsets. Both agree with the published totals at every $m \leq 5$; at $m = 6$ three independent computations (formula, brute force, divisor-structured re-indexing) agree on 688.

(V7) **The cyclotomic engine in exact integer form:** $\prod_{k=0}^{L-1}(1+\zeta_L^k) = 1-(-1)^L$ is the characteristic product of the L-cycle, i.e. the integer determinant $\det(I+C_L) = 1-(-1)^L$ for $L = 1, \ldots, 12$, computed two ways (exact Gaussian elimination; the subset/minor expansion). The sabotage control (an $(L-1)$-cycle plus fixed point) fires with the predicted different value $2(1-(-1)^{L-1})$.

(V8) **The parity wall, finite shadow** (Theorem S3 below): the Gaussian-integer integrality-vs-type solve, and the Künneth convolution computing the E-degree-1 component of products.

(V9) **The canonical projector, finite shadow** (Theorem S1 below).

(V10) **The shadow-theorem arithmetic, finite shadow** (Theorem S2 below): the polynomial-inversion extraction $\hat{\pi} \circ q(\hat{\pi}) = \pi$ computed on random invertible T.

(V11) **The channel composite, finite shadow** (Theorem S5 below): the codimension bookkeeping $(d+p-1)+1+g-(d+1) = p+g-1$ and the action identity on random instances.

(V12) **The canonical duality, finite shadow**: the Poincaré-bundle composite acts as the symplectic duality, squaring to $-I$, nondegenerate.

(V13) **The door law, re-derived this session's own way** (membership computed from the deck definition, cross-checked against direct coordinate solvability; the door-parity: odd scales refused, $m = 2$ admitted).

(V14) **Hostile controls**: the sabotage battery, every control firing with predicted values.

3 Theorem S1: the canonical middle splitting (strengthening Round 32, Theorem 1)

Round 32 defined M_{mid} as the maximal level ≤ 1 rational sub-Hodge-structure of $H^{2p-1}(X, \mathbb{Q})$ and proved it two-level and polarizable. This section strengthens the extraction to a *canonical splitting with an explicit description and a complement-independent projector.*

Theorem 1 (Canonical Middle Splitting)**.** *Let X be a smooth projective complex variety, $p \geq 1$, and write the Lefschetz decomposition of the odd cohomology with respect to a hyperplane class $\mathcal{L}$:*

$$H^{2p-1}(X, \mathbb{Q}) \;=\; \bigoplus_{j \geq 0} \mathcal{L}^j \, P^{\,2p-1-2j}(X, \mathbb{Q}).$$

Then:

(i) ***Explicit description.*** *$M_{\mathrm{mid}} = \bigoplus_j \mathrm{mid}\big(\mathcal{L}^j P^{\,2p-1-2j}\big)$, where $\mathrm{mid}(\cdot)$ denotes the sum of the Hodge-type components with $|a-b| \leq 1$ (the types $(p-j-1, p-j)$ and $(p-j, p-j-1)$ of each primitive piece). In particular M_{mid} is a* canonical direct summand*:*

$$H^{2p-1}(X, \mathbb{Q}) = M_{\mathrm{mid}} \oplus M_{\mathrm{high}}, \qquad M_{\mathrm{high}} := \bigoplus_j \mathrm{high}\big(\mathcal{L}^j P^{\,2p-1-2j}\big),$$

with every Hodge type (a, b) of M_{high} satisfying $|a-b| \geq 3$.

(ii) ***The canonical projector.*** *The* middle projector $\pi_{\mathrm{mid}} \in \mathrm{End}_{\mathbb{Q},\mathrm{Hodge}}\big(H^{2p-1}(X)\big)$ *— projection onto M_{mid} along M_{high} — is* independent of every auxiliary choice*: any two $\mathbb{Q}$-rational Hodge idempotents with image M_{mid} coincide. It is canonically attached to the pair (X, p).*

(iii) ***Nondegeneracy.*** *The polarization pairing of $H^{2p-1}(X)$ restricts nondegenerately to M_{mid}; consequently $M_{\mathrm{high}} = M_{\mathrm{mid}}^{\perp}$ for the standard polarization, and the splitting of (i) is the orthogonal splitting.*

[PROVEN HERE]

Proof. (i) The hyperplane class has type $(1,1)$, so cupping with $\mathcal{L}^j$ shifts a type (a,b) to $(a+j,b+j)$ and preserves $a-b$; the primitive pieces P^k have $P^{a,b} \neq 0$ only for $a+b=k$. For $k = 2p-1-2j$ odd, the types with $|a-b| \leq 1$ are exactly the two middle types $(a,b) \in \{(p-j-1,\, p-j), (p-j,\, p-j-1)\}$. Each summand $\mathrm{mid}(\mathcal{L}^j P^k)$ is a $\mathbb{Q}$-sub-Hodge-structure (its set of types is conjugation-stable, and the Lefschetz decomposition is defined over $\mathbb{Q}$), of level ≤ 1; hence it is contained in M_{mid} by maximality. Conversely, if $v \in M_{\mathrm{mid}}$, decompose $v = \sum_j \mathcal{L}^j v_j$ with v_j primitive; each $\mathcal{L}^j v_j$ is a $\mathbb{Q}$-rational component of a level ≤ 1 class, and since $\mathcal{L}^j$ preserves $a-b$, each v_j's components with $|a-b| \leq 1$ assemble into a sub-Hodge-structure summand of M_{mid}, while the $|a-b| \geq 3$ components assemble into a sub-Hodge-structure of level ≥ 3 contained in M_{mid} — which forces them to vanish (level ≤ 1 has no such types). So v lies in the right-hand side. The complement M_{high} is a sub-Hodge-structure by the same type-stability, and its types satisfy $|a-b| \geq 3$ because $a+b = 2p-1$ is odd, so $|a-b|$ is odd and $\neq 1$.

(ii) Let π, π' be $\mathbb{Q}$-rational Hodge idempotents with image M_{mid}. Then $\pi - \pi'$ vanishes on M_{mid} (both act as the identity there), so it factors through $\ker\pi \cap \ker\pi'$'s ambient complement; precisely, $\pi - \pi' = (\pi - \pi') \circ (1-\pi)$, and $(1-\pi)$ has image M_{high}-typed (any Hodge idempotent with image M_{mid} has kernel with types disjoint from the middle types, by the same sub-Hodge-structure argument as in (i)). A Hodge map sends type (a,b) to type (a,b); since M_{high}'s types and M_{mid}'s types are disjoint, $\mathrm{Hom}_{\mathrm{HS}}(M_{\mathrm{high}}, M_{\mathrm{mid}}) = 0$, so $\pi - \pi' = 0$.

(iii) The polarization pairing S on H^{2p-1} is, on each summand $\mathcal{L}^j P^k$, a nondegenerate pairing of that summand with itself (Hodge–Riemann on the primitive piece, transported by $\mathcal{L}^j$), and distinct Lefschetz summands are orthogonal for S. Within $\mathcal{L}^j P^k$, S pairs type (a,b) against $(k-a, k-b)$ only; for the middle types the pairing partner of $(p-j-1, p-j)$ is $(p-j, p-j-1)$ — both inside the middle sector, and the Hodge–Riemann relations make it nondegenerate there. So $S|_{M_{\mathrm{mid}}}$ is nondegenerate, $M_{\mathrm{mid}} \cap M_{\mathrm{mid}}^{\perp} = 0$, and dimension count gives the orthogonal splitting; $M_{\mathrm{mid}}^{\perp}$'s types are then forced into $|a-b| \geq 3$ by the argument of (i), so $M_{\mathrm{mid}}^{\perp} = M_{\mathrm{high}}$. □

Remark 2 (What S1 changes)**.** *Round 32's Theorem 1 asserted existence and uniqueness of M_{mid} as a maximal sub-*structure*. S1 makes the extraction* functorial-grade*: an explicit Lefschetz-piece formula, a canonical complement, and a canonical projector $\pi_{\mathrm{mid}}(X,p)$ — one named, choice-free $\mathbb{Q}$-rational Hodge idempotent per input. The odd face of the input axis is henceforth the algebraicity question for this one canonically named object (see Theorem S2), not for an unspecified class of projectors.*

4 Theorem S3: the unconditional parity wall at every input

The campaign's parity wall (§42 of the skill, ledger 095) was certified at the committed worlds and at generic splits. Here it is proven *unconditionally, at every input*, in a form this session derived and computed on its own (V8): the E-degree form.

Theorem 3 (Unconditional Parity Wall)**.** *Let E be the Gaussian elliptic curve, X any smooth projective complex variety. Define the* E-degree *of a class in $H^*(E \times X)$ as its Künneth component index in $H^{d_E}(E) \otimes H^{*-d_E}(X)$ (so $d_E \in \{0,1,2\}$; a class is a sum of its E-degree components).*

(i) Every integral $(1,1)$ *class on* $E \times X$ *has E-degree-components only in degrees* 0 *and* 2*: its E-degree-*1 *Künneth component is zero.*

(ii) Consequently every class of the divisor algebra (the $\mathbb{Q}$*-span of finite products of integral* $(1,1)$ *classes) has E-degree-*1 *Künneth component zero.*

(iii) Every nonzero class of the balanced lattice $B = \{x = e \otimes u + f \otimes v : \ u - iv \ \textit{purely}(p-1,p)\}$ *is pure of E-degree* 1*. Hence*

$$B \ \cap \ \{\textit{divisor algebra}\} \ = \ \{0\} \qquad \textit{at} \text{ every } \textit{input } (X,p).$$

[PROVEN HERE]

Proof. (i) Let $c \in H^2(E \times X, \mathbb{Z}) \cap H^{1,1}$, with Künneth components $c = c^{(0)} + c^{(1)} + c^{(2)}$ ($c^{(d)} \in H^d(E,\mathbb{Z}) \otimes H^{2-d}(X,\mathbb{Z})$, each integral). The component $c^{(1)} = e \otimes u + f \otimes v$ with $u, v \in H^1(X,\mathbb{Z})$: integrality means u, v are fixed by conjugation $\overline{(\cdot)}$ on $H^1(X,\mathbb{C})$; the type condition means $e \otimes u$ is of type $(1,1)$, which forces u purely of type $(0,1)$, and $f \otimes v$ of type $(1,1)$ forces v purely $(1,0)$. But a conjugation-fixed vector that is purely $(0,1)$ satisfies $u = \bar{u} \in H^{1,0} \cap H^{0,1} = 0$ (conjugation maps $(0,1)$ to $(1,0)$), and likewise $v = 0$. So $c^{(1)} = 0$. (ii) The Künneth components of a product are convolutions of the components of the factors; a product of classes each with components only in E-degrees 0 and 2 has E-degree-1 component $\sum_{d+d'=1} c^{(d)} c'^{(d')}$ with $d, d' \in \{0,2\}$ — an empty sum. (iii) The balanced $x = e \otimes u + f \otimes v$ lies in $H^1(E) \otimes H^{2p-1}(X)$ by construction: pure E-degree 1; it is nonzero iff $(u,v) \neq (0,0)$. Any class both in B and in the divisor algebra is pure E-degree 1 with vanishing E-degree-1 component, hence zero. □

Remark 4 (The wall's correct scope, and the channel it leaves open)**.** *The wall is a statement about the* divisor algebra *— products of* $(1,1)$ *classes. It is unconditional and input-free. It does* not *close the mixed sector: the two-center conduit operates through* higher Chern classes of rank ≥ 2 bundles*, whose* ch^p *components are pure* (p,p) *but whose Künneth components* do *reach E-degree* 1 *(c_k for $k \geq 2$ has an $H^1(E) \otimes H^{2k-1}(X)$ part, and type-purity forces exactly the balanced middle types). The wall certifies that no supply of the balanced sector can come from divisors; the conduit channel and the correspondence channel (Theorem S5) are the ones that can.* ***[THIS SESSION'S OWN CALCULATION]****: the finite shadow V8 computes the integrality-vs-type solve and the Künneth convolution.*

5 Theorem S5: the divisor-twisted curve-channel supply (unconditional)

This is the round's *construction* theorem — the first unconditional supply of K-correspondences (and hence of algebraic balanced classes, via Round 32's Theorem 2(b⇒a)) beyond $p = 1$ and the committed worlds.

Theorem 5 (Divisor-Twisted Curve-Channel Supply)**.** *Let X be smooth projective, $p \geq 2$, C a smooth projective curve with Jacobian $J = J(C)$ of dimension g, and suppose given:*

- *an algebraic class $\delta \in \mathrm{CH}^{p-1}(X)_{\mathbb{Q}}$ (a divisor-word), and*
- *an algebraic correspondence $V \in \mathrm{CH}^1(X \times C)_{\mathbb{Q}}$.*

Then the composite

$$K_{\delta,V} \;:=\; \Delta_{X,*}\big(\mathrm{cl}(\delta)\big) \;\circ\; V \;\circ\; c_1(\mathcal{P}_C) \;\circ\; \Delta_{J,*}\big(\mathcal{L}_J^{g-1}\big) \;\in\; \mathrm{CH}^{p+g-1}(X\times J)_{\mathbb{Q}}$$

is algebraic, where $\Delta_{X,*}(\mathrm{cl}\,\delta)$ *is the diagonal-correspondence acting as cup-product with* $\mathrm{cl}(\delta)$*,* $\mathcal{P}_C$ *the Poincaré bundle on* $C\times J$*, and* $\mathcal{L}_J$ *a hyperplane class on* J*. Its cohomology action on* $H^1(J)$ *is, up to a nonzero rational scalar (the symplectic normalization), the composite*

$$H^1(J) \;\xrightarrow[\cong]{\text{canonical duality}}\; H^1(C) \;\xrightarrow{V_*}\; H^1(X) \;\xrightarrow{\wedge\,\mathrm{cl}(\delta)}\; H^{2p-1}(X),$$

whose image lies automatically in $M_{\mathrm{mid}}(X,p)$ *(type-chasing:* $V_*\beta$ *has type* $(1,0)$ *or* $(0,1)$*, so the wedge with the* $(p-1,p-1)$ *class* δ *has type* $(p,p-1)$ *or* $(p-1,p)$*). Consequently* $K_{\delta,V}$ *is a* K*-correspondence for the sub-Hodge-structure* $\big[\,\mathrm{cl}(\delta)\wedge V_*H^1(C)\,\big]\subseteq M_{\mathrm{mid}}$*, and every balanced class* x_u *with* u *in this sub-structure is algebraic on* $E\times X$*.* ***[PROVEN HERE]***

Proof. The ingredients are algebraic: δ is algebraic by hypothesis; $\Delta_{X,*}\mathrm{cl}(\delta)\in\mathrm{CH}^{d+p-1}(X\times X)$ (pushforward of a codimension-$(p-1)$ cycle along the codimension-d diagonal) is the correspondence acting as $\wedge\,\mathrm{cl}(\delta)$; V is algebraic by hypothesis; $c_1(\mathcal{P}_C)\in\mathrm{CH}^1(C\times J)$ is a divisor class; $\Delta_{J,*}\mathcal{L}_J^{g-1}\in\mathrm{CH}^{2g-1}(J\times J)$ is the correspondence acting as cup-$\mathcal{L}^{g-1}$ (hard Lefschetz, $H^1\to H^{2g-1}$, both **[STANDARD STRENGTH]**). Composites and sums of algebraic correspondences are algebraic.

The codimension bookkeeping: the composite $[J\to C\to X\to X]$ has codimension

$$\big(d+(p-1)\big)+1+\big(1+(g-1)\big)\;-\;d\;-\;1\;=\;p+g-1,$$

subtracting the intermediate dimensions ($\dim X=d$, $\dim C=1$). **[CERTIFICATE RECEIPT]** (V11 computes it on random instances, and the hostile wrong-bookkeeping control fires.)

The action: on $\beta\in H^1(J)$: cup-$\mathcal{L}^{g-1}$ gives $\mathcal{L}^{g-1}\beta\in H^{2g-1}(J)$; the Poincaré-divisor correspondence $c_1(\mathcal{P}_C)$ contracts it against the canonical duality: writing $\{e_i\},\{f^i\}$ symplectic-dual letter bases of $H^1(C)$ and $H^1(J)$, the $H^1\otimes H^1$ Künneth component of $c_1(\mathcal{P}_C)$ is the canonical tensor, and the contraction sends $\mathcal{L}^{g-1}\beta\mapsto\sum_i e_i\,\langle f^i,\beta\rangle_J$ — the canonical identification $H^1(J)\cong H^1(C)$ up to the symplectic sign (**[STANDARD STRENGTH]**, the defining property of the Poincaré bundle; the finite shadow V12 computes the composite as the symplectic matrix, squaring to $-I$, nondegenerate). Then V_* maps $H^1(C)\to H^1(X)$, and $\wedge\,\mathrm{cl}(\delta)$ lands in $H^{2p-1}(X)$ with the type-chase above placing the image in the middle types. **[CERTIFICATE RECEIPT]** (V11: the composite action equals the channel map on random instances, associativity and evaluation orders independent).

The conclusion: the composite realizes (up to the nonzero rational scalar fixed by the symplectic normalization) the embedding of the sub-structure $\mathrm{cl}(\delta)\wedge V_*H^1(C)$ into $H^{2p-1}(X)$ after the canonical identification $H^1(J)\cong H^1(C)(p-1)$ — which is exactly the K-condition of Round 32's Theorem 2 for that sub-structure (the theorem's proof applies verbatim with M replaced by the sub-structure and J by $J(C)$, both identifications being the ones constructed here). Theorem 2(b⇒a) then transfers $K_{\delta,V}$ into algebraic realizations of the balanced classes x_u for u in the channel. □

Corollary 6 (Products of curves: the (2+1)-sector closes)**.** *Let* $X=C_1\times\cdots\times C_n$ *be a product of smooth projective curves,* $2\le p\le\lfloor d/2\rfloor$*,* $d=n$*. Every Künneth component of* $H^{2p-1}(X)$ *of type* $\big[H^2(C_i)\otimes H^1(C_j)\big]$ *(a "2+1" component, with the remaining* $p-2$ *curve factors carrying* H^2*-divisor content and the rest* H^0*) lies in* M_{mid} *and is* K*-supplied by Theorem* S5 *with* $V=\Gamma_{\pi_j}$ *the graph of the projection (a codimension-1 correspondence, acting as* π_j^* *on* H^1*) and* δ *the divisor-word. Hence*

the entire (2+1)*-sector of* M_{mid} *at products of curves is* K*-certified and its balanced classes are algebraic. The* residual *middle content at products of curves is exactly the pure odd-tensor sector: the level* ≤ 1 *parts of* $\bigotimes_{i\in S} H^1(C_i)$ *over* $|S| = 2p-1$ *(for the first open cell* $d = 4, p = 2$*: the middle part of* $H^1 \otimes H^1 \otimes H^1$*). **[PROVEN HERE]***

Proof. The (2+1) Künneth component with $\beta \in H^1(C_j)$ and divisor-word δ is exactly $\pi_j^*\beta \wedge \delta = V_*\beta \wedge \delta$ with $V = \Gamma_{\pi_j} \in \mathrm{CH}^1(X \times C_j)$ (the graph of a morphism to a curve is a codimension-1 class acting by pullback). Its types are $(1,0),(0,1)$ tensored with $(p-1,p-1)$ — middle types, so it lies in M_{mid}; Theorem S5 supplies $K_{\delta,V}$. For the residual claim: the Künneth decomposition of $H^{2p-1}(\prod C_i)$ is the direct sum over degree-distributions $k_i \in \{0,1,2\}$ with $\sum k_i = 2p-1$; the middle-typed parts of the distributions with all $k_i \leq 1$ are precisely the pure odd-tensor sector $\bigotimes_{i\in S} H^1$ with $|S| = 2p-1$; the distributions with some $k_i = 2$ are the (2+1)-type components (two H^2's would force $\sum \geq 2p$ unless $p = 1$, the closed sector). The middle-typed part of $\bigotimes_{i\in S} H^1(C_i)$ has no divisor factor to absorb into δ, so no channel of Theorem S5 reaches it. □

Remark 7 (Fibrations)**.** *The same theorem applies at* any X *with a morphism* $\pi\colon X \to C$ *to a curve (*$V = \Gamma_\pi \in \mathrm{CH}^1(X \times C)$ *acts as* π^**): every divisor-twisted sector* $\delta \wedge \pi^* H^1(C) \subseteq M_{\mathrm{mid}}$ *is* K*-supplied, unconditionally. This is the trajectory change on the supply side: a family of channels that the campaign had never struck, certified alive at every input they reach, by explicit composites of classical algebraic objects (graphs, Poincaré divisors, Lefschetz powers, diagonals).*

6 Theorems S2 and S4: the shadow bridge and the abelian landing

Theorem 8 (The Lefschetz Bridge; conditional form, honestly labeled)**.** *Let* (X,p) *be an input of the open sector and suppose the* K*-correspondence exists at* (X,p) *(Round 32, Theorem 2(b)). Suppose further that the* Lefschetz adjoint operator at degree $2p-1$ *is algebraic at* X *— i.e. that there exists an algebraic correspondence* $*_X \in \mathrm{CH}^d(X \times X)_{\mathbb{Q}}$ *acting on* $H^{2p-1}(X)$ *as the adjoint of cup-*$\mathcal{L}^{\,d-4p+2}$ *(in the hard range* $d \geq 4p-2$*; in the range* $d < 4p-2$ *the same statement with the adjoint of the* $\mathcal{L}^{\,4p-2-d}$*-transfer on the Poincaré-dual degree). Then the canonical middle projector* $\pi_{\mathrm{mid}}(X,p)$ *of Theorem* S1 *is algebraic: realized by an explicit polynomial in the algebraic correspondence*

$$\hat{\pi} \;:=\; K \circ \Big(*_X \circ K^t \circ \Delta_{J,*}\mathcal{L}_J^{\,g-1} \Big) \;\in\; \mathrm{CH}^d(X \times X)_{\mathbb{Q}},$$

namely $\pi_{\mathrm{mid}} = \hat{\pi} \circ q(\hat{\pi})$ *where* q *is the minimal-polynomial inverse of the invertible part of* $\hat{\pi}$*'s action on* M_{mid}*. **[CONDITIONAL]** (the* K*-hypothesis is the standing open condition; the* $*_X$*-hypothesis is an instance of the Lefschetz standard conjecture at degree* $2p-1$*; the extraction arithmetic is unconditional)*

Proof. The composite $K^\ddagger := *_X \circ K^t \circ \Delta_{J,*}\mathcal{L}_J^{g-1}$ is an algebraic correspondence $[X \to J]$ whose cohomology action is the adjoint of K_* with respect to the hard-Lefschetz pairings $\langle u, v\rangle_X = \int \mathcal{L}^{d-4p+2} uv$ (hard range) and $\langle \beta, \beta'\rangle_J = \int \mathcal{L}^{g-1}\beta\beta'$: the transpose-with-adjoint identity for correspondences, each factor algebraic by hypothesis. Then $\hat{\pi} = K \circ K^\ddagger$ acts on $H^{2p-1}(X)$ as $\iota \circ \iota^\ddagger$ where $\iota\colon H^1(J) \cong M(p-1) \hookrightarrow H^{2p-1}(X)$ is the identification: this action has image in $M(p-1)$, kills the pairing-orthogonal complement of $M(p-1)$, and restricts on $M(p-1)$ to an *invertible* endomorphism T (the adjoint composite of an injective map between nondegenerately paired spaces is invertible on the image). So $\hat{\pi}$ acts as $T \circ \pi'$ with π' a projector onto $M(p-1)$-along-a-complement; by Theorem S1(ii) the Hodge-projector with image M is unique, so $\pi' = \pi_{\mathrm{mid}}$ after the twist bookkeeping, i.e. $\hat{\pi}$ acts as $T\pi_{\mathrm{mid}}$ with $T \in \mathrm{End}_{\mathrm{HS}}(M(p-1))$ invertible. Let $m(x) = c_0 + c_1 x + \cdots$ be the minimal polynomial of T ($c_0 \neq 0$ by invertibility) and $q(x) := -\big(c_{r-1} + \cdots\big)/c_0$ the polynomial

with $q(T)T = 1 - \frac{m(T)}{c_0}c_0^{-1} \cdot c_0 = 1$ on the spectrum, i.e. $q(T) = T^{-1}$ and $q(0) = 0$. Since $\hat{\pi}^k$ acts as $T^k\pi_{\mathrm{mid}}$, $q(\hat{\pi})$ acts as $q(T)\pi_{\mathrm{mid}}$, and

$$\hat{\pi} \circ q(\hat{\pi}) \text{ acts as } T\pi \cdot T^{-1}\pi = \pi_{\mathrm{mid}},$$

an algebraic correspondence (a polynomial in $\hat{\pi}$) realizing the canonical projector. **[CERTIFICATE RECEIPT]** (V10: the polynomial-inversion extraction computed on random invertible T, $\hat{\pi} \circ q(\hat{\pi}) = \pi$ exactly). □

Remark 9 (What S2 establishes, and what it honestly costs)**.** *Unconditionally, S2 connects the campaign's K-condition to the classical conjecture network: K* at (X, p) + the Lefschetz standard conjecture at $(X, 2p-1) \Rightarrow$ the canonical middle projector is algebraic. *The extraction itself (the polynomial inversion) is unconditional arithmetic, certified. The $*_X$-hypothesis is an instance of the Lefschetz standard conjecture (Grothendieck), open in general and known in the campaign's committed classes; this is stated on the theorem's face, at **[CONDITIONAL]** strength — no unconditional projector-algebraicity is claimed at general input, and none should be: the type of operator required genuinely lives behind that conjecture (the degree bookkeeping forces an inverse-Lefschetz-shaped correspondence; the campaign's own certificate V11 degree-chase records the same obstruction on the composition side).*

Theorem 10 (The Abelian Landing)**.** *Let $X = A$ be an abelian variety and $p \geq 2$ an open-sector degree. Then the following are equivalent:*

(a) the K-correspondence exists at (A, p);

(b) the mixed closure holds at (A, p): every balanced class $x \in B \subseteq H^{2p}(E \times A, \mathbb{Q})$ is algebraic;

(c) every rational (p, p) class on the abelian variety *$E \times A$ that lies in the balanced lattice B is algebraic.*

Moreover the balanced sector is parity-wall-clean: by Theorem S3*, B meets the divisor algebra of $E \times A$ trivially — so the odd face of the input axis at abelian X is exactly a* beyond-divisor *sector of the Hodge conjecture for the abelian variety $E \times A$ (the classical territory of Weil classes on abelian varieties of Weil type and their generalizations). And by Theorem* S2 *(conditional form), $K \Rightarrow \pi_{\mathrm{mid}}(A, p)$ algebraic modulo the Lefschetz adjoint, which for abelian A is a classical instance. **[PROVEN HERE]** (the equivalence is Round 32's Theorem 2 at $X = A$; the wall is Theorem S3; nothing is claimed about the abelian Hodge conjecture itself)*

Proof. (a)⇔(b) is Round 32's Theorem 2. (b)⇔(c): the balanced classes are exactly the rational (p, p) classes in $\Lambda_{\mathrm{mix}} \otimes \mathbb{Q}$ with the balanced type condition — they are (p, p) classes on $E \times A$, which is an abelian variety (product of abelian varieties), and conversely a rational (p, p) class lying in the balanced lattice is balanced by definition. The parity wall is Theorem S3 at $X = A$. The last clause is Theorem S2 with $X = A$. □

7 The final phase state: the input axis re-drawn

Theorem 11 (The Strengthened Terminal Map)**.** *The rational Hodge conjecture on the open sector $2 \leq p \leq \lfloor d/2 \rfloor$, $d \geq 4$, is equivalent to [K-correspondences everywhere + transcendental algebraicity] (Round 32, Theorem 4 — unchanged, and still the equivalence-strength boundary). The sector map of the input axis is now:*

Sector / channel	***Content***	***Status***	***Receipt***
Divisor algebra (all inputs)	*products of* $(1,1)$	***closed, and certified DEAD as a supply for the balanced sector at every input***	*S3* ***[THIS SESSION'S OWN CALCULATION]***
Mixed, $p = 1$	*Albanese graph*	*closed*	*R32 Cor.*
Curve channels (all inputs they reach)	$\delta \wedge V_* H^1(C)$*-sectors*	***closed — NEW, unconditional***	*S5* ***[THIS SESSION'S OWN CALCULATION]***
Products of curves, (2+1)*-sector*	*Künneth* $H^2 \otimes H^1$ *parts*	***closed — NEW***	*S5 Cor.*
Committed worlds	*census rows,* $\{S_m(A)\}$	*closed*	*census, R32*
Abelian X*, odd face*	*balanced* (p,p) *on* $E \times A$	***localized to the classical beyond-divisor abelian sector***	*S4*
General X*, projector shadow*	$\pi_{\mathrm{mid}}(X,p)$	***bridged:*** $K \Rightarrow$ ***algebraic mod. the Lefschetz adjoint***	*S2* ***[CONDITIONAL]***
Transcendental face	*fold-free core*	*irreducible*	*R32 Thm.4*

The named residual, after this round: ***(R1)*** *the pure odd-tensor middle parts at products of curves (first open cell: the middle of* $H^1 \otimes H^1 \otimes H^1$ *at* $d = 4, p = 2$*);* ***(R2)*** *the beyond-divisor balanced sector on* $E \times A$ *at abelian* X*;* ***(R3)*** *the* K*-condition at general* (X,p) *beyond the channels;* ***(R4)*** *the transcendental fold-birth (items (i)–(iii) of the standing reduction with the* (p,p)*-sensitive gate).* Q1 *remains proven and certified;* Q2 *is not claimed; the closing string is not emitted.* ***[PROVEN HERE]*** *(as a map: each row's status is proven or honestly labeled)*

The quantifier line (§37)

What this round did to the universal quantifier: the input axis's channel structure was re-drawn from one boundary into four named channel classes — one certified dead at every input (the divisor channel, unconditionally), one family certified *alive and closing* at every input it reaches (the divisor-twisted curve channels, unconditionally), one localized to a classical named sector (abelian), one bridged to the standard-conjecture network (the projector shadow) — with the residual universal content pinned to the four named objects above. The Clay statement is not claimed; the closing string is not emitted.

The honest scope

Nothing in this round discharges (R1)–(R4) at general input. What changed is the *trajectory*: the law stack is strictly stronger (S1: canonical splitting vs. sub-structure; S3: every input vs. committed worlds; S5: an unconditional supply family where none existed; S4–S2: two new bridges to classical territory), every falsity claim the campaign ever made is now backed by this session's own computations (V1–V5), and the boss's own arithmetic (the arity totals, the cyclotomic engine) is re-certified by two fresh algorithms (V6–V7). The falsifiable line, sharpened: construct a channel-realizing supply for one pure odd-tensor middle part (R1), or one beyond-divisor balanced class on one $E \times A$ (R2), or the K-condition at one general input beyond the channels (R3), or the fold-birth at one transcendental input with the input-sensitivity gate firing (R4) — any one of these is a complete, load-bearing, auditable round.

8 The certificate

`trajectory_verification_round33_checks.py`: 844 checks, exit 0, deterministic double-run byte-identical (md5 14be18d7566307dae8ab8127da6e6f7e0), exact arithmetic ($\mathbb{Z}$, $\mathbb{Q}$ via `fractions.Fraction`, Gaussian integers as pairs), zero floats, matrices $\leq 12 \times 12$, brute-force enumerations $\leq 2^{12}$, random instances ≤ 400 per test, runtime < 60 s, hostile controls all firing with predicted values, and the vacuity scan (the ledger-092/107 patterns) clean on the source. The certificate's own development caught and repaired five defects in this session's first drafts (a broken test-matrix construction, a wrong hardcoded expectation, an f-string bug, one self-comparison tautology, one mis-stated parity claim) — the audit discipline applied to this session's own code before any claim was made on it.